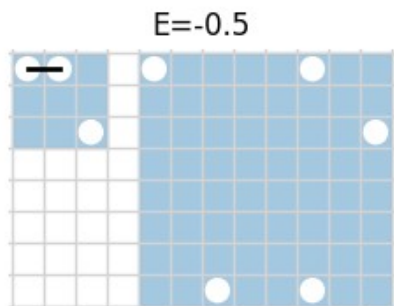


A small system in contact with a heat bath

An exercise of 8 particles in a compound system, consisting of a small system and a heat bath: A 3x3 and an 8x8 lattice.



Each system connects to an earlier exercise: The 3x3 lattice had 2 particles, the 8x8 lattice had 6 particles.

To study how a small system behaves when in contact with a heat bath, we wish to compare it with the earlier exercise.

First, we determine the canonical probability of each microstate of the two particles in a 3x3 lattice and plot the probability of observing each energy (-0.5, -0.25, 0).

Now we put the small system together with the large system which acts as a finite heat bath. We wish to sample the microstates in the compound system with 2 particles in the smaller subsystem, leaving 6 particles in the finite heat bath. In the larger system we found $\langle E_{bath} \rangle \approx -1.7$ at $k_B T = 0.4$. The canonical average energy of the subsystem at this temperature is $\langle E_{sub} \rangle \approx -0.3$. Together, we need $\langle E_{compound} \rangle = \langle E_{sub} \rangle + \langle E_{bath} \rangle \approx -2$ to achieve $k_B T = 0.4$ for the compound system. These are our criteria for the rejection sampling.

After sampling, we count the microstates of each energy (-0.5, -0.25, 0.0) and normalize by all sampled microstates. We find that it almost exactly follows the canonical distribution, without making any assumptions of it.

